

Week 5 Peer Review

Anonymous Reviewer

I. TITLE

CVaR-MPPI for Highway Lane-Change Planning

II. SUMMARY

The report proposes a risk-sensitive highway lane-change planner that augments MPPI [1] with a CVaR objective [2]. Initial experiments compare against risk-neutral MPPI in HighwayEnv [?]. The baseline avoids crashes but violates comfort and safety limits, with very large lateral acceleration, jerk, and low TTC.

III. NOVELTY

The idea is well motivated, but the novelty should be stated more explicitly relative to Risk-Aware MPPI [5]. Since CVaR-MPPI has already been proposed, the report should clarify whether the main contribution is the highway lane-change formulation, the use of HighwayEnv, the offline CVaR-MDP terminal value, or the specific safety/comfort evaluation.

IV. EVALUATION OF SCOPE

The project comprises an extensive study that includes online CVaR-MPPI, stochastic traffic prediction, JAX acceleration, and offline CVaR-MDP value iteration. A focused final project could prioritize online CVaR-MPPI and compare it carefully with risk-neutral MPPI. The offline value function from CVaR-MDP theory [3], [4] can be treated as an extension if time permits.

V. SUGGESTIONS FOR METHOD

The authors should clarify how surrounding vehicles are modeled. Lane changing is a multi-agent interaction problem, but the current formulation appears to treat traffic mainly as stochastic exogenous dynamics. Although HighwayEnv uses IDM/MOBIL, it is unclear whether the MPPI traffic prediction reacts to each sampled ego trajectory or only adds independent disturbances. This matters because the CVaR tail estimate should reflect realistic interactive risk.

The CVaR computation should also be specified more clearly, following Risk-Aware MPPI [5]: the loss $L(\tilde{x})$, empirical VaR calculation, number of stochastic rollouts, and penalty weight should be defined. A concise ablation between risk-neutral MPPI, online CVaR-MPPI, and CVaR-MPPI with MDP terminal value would help identify which component gives the improvement.

VI. SUGGESTIONS FOR EXAMPLE

The hard-brake and cut-in cases are good initial tests. To better reflect lane changing as a multi-agent problem, the authors could add scenarios where surrounding vehicles react to ego behavior, such as blocking a merge, yielding, or cutting in after ego starts changing lanes. The final evaluation should use multiple random seeds and report collision rate, near-miss rate, minimum TTC, lateral acceleration, jerk, lane-change duration, and success rate. A case where the safest action is to delay or reject the lane change would also be useful.

VII. SUGGESTIONS ON WRITING

The writing is clear, but it should more explicitly separate completed work from planned work. In particular, the current results appear to use only risk-neutral MPPI, while the offline CVaR-MDP terminal cost is still a future step. The report should also define the CVaR loss $L(\tilde{x})$ more explicitly. There is also a notation inconsistency: Fig. 1 refers to $M = 128$ sampled control sequences, while the baseline description states $M = 256$. The authors should clarify whether these are different experiments or a typo.

VIII. ADDITIONAL COMMENTS

Overall, this is a strong direction. The baseline results support the motivation for CVaR because average-cost MPPI can avoid collisions while still producing risky tail behavior. The final report would be stronger if it shows that CVaR reduces near-misses and unsafe TTC, while still meeting the real-time control budget.

REFERENCES

- [1] G. Williams, A. Aldrich, and E. A. Theodorou, "Model predictive path integral control: From theory to parallel computation," *Journal of Guidance, Control, and Dynamics*, vol. 40, no. 2, pp. 344–357, 2017.
- [2] R. T. Rockafellar and S. Uryasev, "Optimization of conditional value-at-risk," *Journal of Risk*, vol. 2, no. 3, pp. 21–41, 2000.
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- [4] Y. Chow, A. Tamar, S. Mannor, and M. Pavone, "Risk-sensitive and robust decision-making: a CVaR optimization approach," in *Advances in Neural Information Processing Systems*, vol. 28, 2015.
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